HARIS MD SAHED

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Tokyo, Japan

Research Interests

Machine learning under nonstationarity and class imbalance; human-in-the-loop learning; diagnostic-driven hybrid modeling; anomaly and rare-event detection in high-frequency data; applied statistics for decision-making; interpretable and robust AI systems.

Education

Tokyo International University Tokyo, Japan
Bachelor of Science in Digital Business & Innovation (AI / Data Science Track)
Apr 2022 – Mar 2026 (Expected)
GPA: 3.741 / 4.00

Core Coursework:

- Fundamentals of Mathematics
- Foundations of Python
- Mathematics for Business and Economics I
- Introduction to R Programming
- Statistics I & II
- Big Data and Analytics
- Introduction to Informatics & Computing
- Introduction to Computer Networking
- AI & Machine Learning for Decision Making
- Text Mining & Deep Learning
- Introduction to AI & Intelligent Product Development
- Creative Thinking for AI & IoT Innovations
- Blockchain & Business Applications
- Quantum Information and Computation

Undergraduate Thesis

A Diagnostic-Driven Framework for Selecting Hybrid ML Strategies in Nonstationary and Imbalanced Data
Undergraduate Thesis Dec 2025
Advisor: Prof. Fatih Ozaydin, Tokyo International University

- Developed a diagnostic-driven framework for selecting hybrid learning strategies under nonstationarity and class imbalance.
- Demonstrated limitations of fixed pipelines in rare-event and distribution-shift scenarios.
- Improved robustness and balanced accuracy through diagnostic-informed model and resampling selection.

Research Experience

Undergraduate Researcher 2024 – 2025
Supervisor: Prof. Fatih Ozaydin, Tokyo International University

- Conducted research on extreme climate event forecasting using machine learning and deep learning models. Addressed challenges including class imbalance, nonstationarity, low signal-to-noise ratios, asynchronous data, and intraday seasonality.
- **Ongoing research:** Developing a diagnostic-triggered Human-in-the-Loop (HITL) machine learning framework. Designing statistical and model-based diagnostics to trigger selective human intervention during concept drift and high uncertainty.

Student Assistant (Statistics) 2023 – 2024
Prof. Makoto Shimizu, Tokyo International University

- Supported grading and verification of assignments involving descriptive statistics and basic inferential methods.

Publications

1. **Haris Md Sahed**, Md Abu Saleh Showrob, Fatih Ozaydin. *Evaluation of Machine Learning and Deep Learning Models for Extreme Climate Event Forecasting.* In: *Proceedings of the 2025 International Conference on Advanced Machine Learning and Data Science (AMLDS)*, Tokyo, Japan, 19–21 July 2025. IEEE. DOI: 10.1109/AMLDS63918.2025.11159347

Selected Projects

Traffic Congestion & Smart City Planning 2024
AI & Machine Learning for Decision Making

- Developed predictive models using IoT traffic data for congestion and accident detection.
- Applied XGBoost, LightGBM, and Prophet for time-series forecasting and pattern analysis.

Bayesian Learning in Text Mining and Deep Learning 2025

- Compared Naive Bayes, LSTM, and Bayesian Neural Networks on sentiment analysis tasks.
- Demonstrated improved reliability via uncertainty-aware Bayesian modeling.

Awards and Honors

- JASSO Scholarship for Privately Financed International Students (2022–2023)
- Soka City Hall Scholarship for Undergraduate Students (2023–2026)

Technical Skills

Programming & Data Science: Python, R, SQL, scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch
Statistics & Tools: Hypothesis testing, regression, time-series analysis, LaTeX, Tableau

Languages

English (Fluent — TOEFL iBT MyBest: 97), Japanese (Intermediate), Bengali (Native)